

SHIH-WEN LIU (CASPER)

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RESEARCH INTERESTS

My core interests are **efficient and adaptive AI for multimodal and multi-task learning**. I particularly focus on parameter-efficient fine-tuning, adaptive mechanisms for multi-task learning vision model, large multimodal model and histopathology report generation. My research goal is to develop AI systems that can efficiently learn, adapt, and transfer knowledge across diverse tasks and modalities.

EDUCATION

- **National Cheng Kung University** Sep. 2024 – Present
Tainan, Taiwan
Master of Science in Artificial Intelligence (Advisor: Prof. Wei-Ta Chu)
 - CGPA: 4.00/4.00
 - Thesis: *Frequency-Switching Mechanism for Parameter-Efficient Multi-Task Learning*
- **National Sun Yat-sen University** Sep. 2020 – Jun. 2024
Kaohsiung, Taiwan
Bachelor of Science in Computer Science Engineering
 - CGPA: 3.62/4.00, Academic Excellence Award (2020)

PUBLICATIONS

- [C.1] **Shih-Wen Liu, Yen-Chang Chen, Wei-Ta Chu, Fu-En Yang, Yu-Chiang Frank Wang** "**Frequency Switching Mechanism for Parameter-Efficient Multi-Task Learning**." In *Proceedings of the IEEE/CVF International Conference on Computer Vision (CVPR) 2026*. (Work done in collaboration with NVIDIA.)
- [C.2] **Shih-Wen Liu, Hsuan-Yu Fan, Wei-Ta Chu, Fu-En Yang, Yu-Chiang Frank Wang** "**Histopathology Image Report Generation by Vision Language Models with Multimodal In-Context Learning**." In *the Conference of Medical Imaging with Deep Learning (MIDL) 2025*. (Work done in collaboration with NVIDIA.)

PROJECTS

- **MARL: Multi-Agent Reinforcement Learning for Adhoc Teamwork** Jan. 2026
Tools: Python, PyTorch
 - Developing a collaborative robotic platform integrating perception, decision-making, and human interaction.
 - Exploring reinforcement learning and foundation-model-based reasoning for adaptive human-robot cooperation.
 - Building real-world robotic systems using ROS2 and Unity simulation.
- **SUIII! A Ball Focus and Identity Memory for Robust Soccer Player Tracking** Feb. 2025
[Project]
Tools: Python, PyTorch
 - Improved soccer player tracking under occlusion using ball-focused reasoning, group motion dynamics, and long-term appearance memory

EXPERIENCE

- **Research Collaborator, NVIDIA Research Taiwan** [🌐] Sep. 2024 – Jan. 2026
Taiwan
Mentors: Dr. Fu-En Yang, Prof. Yu-Chiang Frank Wang
 - Collaborated on parameter-efficient multi-task learning and multimodal models.
 - Co-authored publications at MIDL 2025 and CVPR 2026.
- **Intern, NXP Semiconductors** [🌐] Feb. 2023 – Aug. 2023
Kaohsiung, Taiwan
Mentor: Yolanda Chen
 - Developed automated data preprocessing tools, algorithms, and logic validation scripts for large-scale data.
 - Enhanced existing pipelines to improve performance and maintainability.

TEACHING

- **Multimedia Analysis in Sport Technology, NCKU CSIE** [🌐] Spring 2026
Tainan, Taiwan
Teaching Assistant
 - Assisted in courses and assignments design.
- **Introduction to Artificial Intelligence, TAICA** [🌐] Fall 2025
Tainan, Taiwan
Teaching Assistant
 - Selected course under the TAICA Program with students from 50+ universities.
 - Assisted in courses and assignments design.
- **Linear Algebra, NCKU CSIE** [🌐] Spring 2025
Tainan, Taiwan
Teaching Assistant
 - Assisted in courses and graded assignments.
- **Scratch Programming** Summer 2021 & 2022
Hsinchu, Taiwan
Instructor
 - Taught introductory Scratch programming to elementary students, boosting communication and teaching skills.

HONORS AND AWARDS

BAIR Summer Grant, NCKU School of Computing, 2026.
Student Travel Grant, National Science and Technology Council, 2025.
Best Team Award, Taiwan Datathon, 2024.

SKILLS

Programming Languages	Python, C/C++, JavaScript, HTML, CSS
Machine Learning Packages	NumPy, Pandas, PyTorch, JAX, TensorFlow, scikit-learn, Matplotlib
Developer Tools	Docker, ROS2, Unity, React Native, Git, Bash, App Development
Languages	English (Fluent), Mandarin (Native), Japanese (Beginner)
TOEFL iBT	99/120 (R:25, L:27, S:23, W:24)